

Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher and software engineer specializing in **Autonomous Decision-Making**, **Spatial AI**, and **Cyber-Physical Systems**. Focused on developing deterministic state frameworks and intelligent agent architectures under tight real-world constraints.

Education

University of California, Santa Barbara (UCSB)

M.S. in Computer Science (GPA: 3.91/4.00)

Graduate Research focused on Spatial AI, AR/VR, and HCI at the Four Eyes Lab.

Advisor: [Prof. Tobias Höllerer](#) · Co-Advisor: [Prof. Michael Beyeler](#)

Santa Barbara, CA

Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Graduated with First Class Honors.

Kharagpur, India

Jul 2016 – Jul 2020

Skills

Programming Languages (Expert): Python, C++, Kotlin, Java, MATLAB

Frameworks & Core Tools (Advanced): Android/iOS Native SDKs, LangChain, Pydantic (JSON Schema), OpenCV, ROS (Robot Operating System), PyTorch

Methodologies & Core Research: Autonomous Decision-Making, Human-Computer Interaction (HCI), Spatial AI, Agentic Architectures, Multi-Modal Sensor Fusion, Finite State Automation

Publications

Refereed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Proceedings of the 14th International Conference on COMmunication Systems & NETworkS (COMSNETS), 2022.

DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Created inertial tracking methods to decode architectural hand stroke motions into alphanumeric symbols on wrist-worn wearables.

Refereed Poster Papers

Eye Tracking Performance in Mobile Mixed Reality

IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2024.

DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

Investigated low-latency mobile eye tracking performance profiles across modern mixed reality layouts. Supported by NSF Grant #2211784.

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), 2023.

DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Proposed unique calibration logic pipelines specifically optimized for eye tracking modalities within mixed-reality locomotion patterns. Supported by NSF Grant #2211784.

Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Projects

JitterNot

Jan 2022 – Mar 2022

LSTM-Based Model Predictive Controller for Adaptive Video Streaming

UCSB

- Developed an LSTM network for real-time throughput prediction, transforming the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

Autonomous Ground Vehicle & Swarm Robotics

Jan 2016 – Dec 2018

End-to-End Perception, Planning, and Multi-Agent Navigation Stack

IIT Kharagpur

- Built an autonomous navigation stack in ROS, fusing heterogeneous sensor streams (LiDAR, IMU, GPS) via an Extended Kalman Filter (EKF).
- Implemented visual obstacle segmentation networks (U-Net) along with global (A^* /Dijkstra) and local (DWA) path planners.
- Researched decentralized swarm coordination using peer-to-peer relative pose estimation (AprilTags) over an ad-hoc XBee mesh network.

In Motion — HAR using DeepConvLSTM

Nov 2019 – Nov 2019

Distributed Mobile Computing System for Human Activity Recognition

IIT Kharagpur

- Designed a client-server telemetry pipeline streaming high-frequency IMU sensor arrays for rolling window inference.
- Implemented a DeepConvLSTM core architecture using TimeDistributed CNN layers to model spatial-temporal features efficiently.

CohesiveAR

Apr 2022 – Jun 2022

Augmented Reality Pipeline for Texture Extraction and Homography Mapping

UCSB

- Engineered an asset ingestion pipeline using Google ARCore and OpenCV to project 3D world coordinates into 2D display spaces.
- Implemented perspective warp and homography transformations to rectify camera tilt skew into standardized orthogonal textures.

Tweeter: Distributed Infrastructure Testbed

Sep 2021 – Dec 2021

High-Concurrency Microblogging Testbed on AWS EC2 Clusters

UCSB

- Conducted high-throughput load testing up to 512 users/sec, mapping user trajectories as state machines to latency scaling curves.

- Optimized read throughput by deploying nested fragment caches and database query preloading to eliminate bottleneck points.

Teaching

University of California, Santa Barbara

Graduate Teaching Assistant

Santa Barbara, CA

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Academic Service & Peer Review

Technical Program Committee & Peer Reviewer

ACM Symposium on User Interface Software and Technology (UIST)

May 2024

Served as an external peer reviewer evaluating full paper submissions for novel interface hardware, interactive system architectures, and spatial tracking methodologies.